

KALISZ, Poland

WMO: 124350

Lat: 51.7812N

Lon: 18.0810E

Elev: 453

StdP: 14.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.5	12.8	2.7	6.4	8.6	7.3	8.1	13.6	27.9	40.6	25.3	40.1	6.7	60	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	86.8	68.5	83.2	67.2	80.1	65.9	70.7	82.3	69.0	80.1	67.5	77.2	7.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	101.1	75.0	65.4	95.3	73.0	63.9	90.5	71.7	34.8	82.6	33.4	80.2	32.2	77.3	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.3	19.1	16.7	3.6	92.8	7.4	2.8	-1.7	94.8	-6.0	96.5	-10.2	98.1	-15.5	100.1		
			3.0	73.7	7.2	1.6	-2.2	74.8	-6.4	75.7	-10.4	76.6	-15.6	77.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	30.0	32.6	38.7	49.0	57.1	62.9	67.0	66.6	58.2	48.9	40.2	32.9
	DBStd	15.06	9.38	8.81	7.87	7.84	7.05	6.49	6.08	5.75	6.29	7.31	7.30	8.55
	HDD50	2534	620	488	355	110	16	0	0	0	6	111	298	530
	HDD65	6215	1085	908	814	482	259	115	48	48	220	499	743	994
	CDD50	2089	0	0	6	80	237	388	528	516	251	77	6	0
	CDD65	297	0	0	0	2	16	53	111	100	15	0	0	0
	CDH74	2612	0	0	0	30	159	478	1006	813	124	2	0	0
	CDH80	722	0	0	0	3	22	131	312	227	27	0	0	0
	WSAvg	8.2	9.4	9.3	9.3	8.4	7.6	7.2	7.2	6.9	7.4	8.0	8.4	9.3
	PrecAvg	20.0	1.1	1.0	1.3	1.1	2.1	2.2	3.2	2.1	1.8	1.3	1.3	1.2
Precipitation	PrecMax	26.7	2.2	2.0	3.1	3.3	5.9	4.1	6.4	5.9	3.9	3.5	4.3	3.2
	PrecMin	11.3	0.1	0.1	0.5	0.1	0.7	0.6	1.0	0.4	0.5	0.2	0.0	0.4
	PrecStd	3.5	0.6	0.5	0.7	0.7	1.1	1.1	1.5	1.1	1.0	0.8	0.7	0.7
	PrecStd	3.5	0.6	0.5	0.7	0.7	1.1	1.1	1.5	1.1	1.0	0.8	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.7	53.9	63.5	78.2	83.1	89.7	91.9	90.8	83.6	72.4	59.3	52.3
		MCWB	47.2	47.5	52.4	60.5	65.8	70.7	69.6	69.3	65.5	61.4	53.3	49.1
	2%	DB	47.0	50.0	58.1	71.5	78.6	83.5	87.1	85.6	77.2	67.3	55.8	48.8
		MCWB	44.2	45.5	49.5	56.9	64.2	67.7	68.7	68.1	63.4	58.9	51.8	46.1
	5%	DB	44.3	46.9	54.0	66.4	74.6	79.3	83.3	82.0	73.1	63.5	52.8	46.4
		MCWB	42.0	43.6	47.6	54.7	62.2	65.4	67.3	67.2	61.5	57.1	49.6	44.3
	10%	DB	41.5	43.9	50.1	62.3	70.7	75.5	79.8	78.5	69.4	59.6	50.1	43.5
		MCWB	39.7	41.1	45.2	52.4	59.8	63.5	66.2	65.5	60.3	55.2	47.6	41.6
	0.4%	WB	48.1	48.9	53.4	61.8	68.7	72.2	73.5	72.1	67.4	63.3	54.9	49.8
		MCDB	50.3	52.5	61.6	74.8	79.5	84.6	84.3	83.0	79.0	69.5	58.1	51.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.5	46.4	50.5	58.6	65.6	69.3	70.9	70.1	65.0	59.8	52.2	46.6
		MCDB	46.5	49.2	56.5	69.5	75.1	80.9	82.5	82.0	74.6	65.4	54.7	48.4
	5%	WB	42.1	43.7	48.1	55.7	63.4	67.1	69.1	68.6	63.0	57.7	49.9	44.3
		MCDB	44.1	46.6	53.0	64.6	72.5	76.4	79.8	79.4	70.6	62.3	52.6	46.2
	10%	WB	39.7	41.2	45.7	53.3	61.1	64.9	67.5	67.0	61.0	55.6	47.7	41.8
		MCDB	41.5	43.8	49.7	60.6	69.3	73.1	77.2	76.1	67.4	59.5	49.8	43.4
Mean Daily Temperature Range	5% DB	MDBR	7.6	9.3	12.2	16.2	17.1	17.2	17.7	17.7	15.3	12.4	8.6	7.3
		MCDBR	9.3	12.9	18.9	22.6	22.8	23.2	23.5	23.1	21.7	18.0	12.6	9.2
		MCWBR	7.8	9.5	12.0	11.9	11.4	10.3	8.7	9.6	9.8	10.6	9.2	7.6
	5% WB	MCDBR	9.1	12.2	17.3	20.5	20.8	20.7	20.5	20.6	18.6	16.4	11.9	9.1
		MCWBR	8.0	9.5	11.7	11.5	11.5	10.7	9.2	9.7	9.6	10.3	9.1	7.8
Clear-Sky Solar Irradiance	taub		0.316	0.342	0.380	0.426	0.420	0.415	0.444	0.431	0.395	0.369	0.345	0.313
	taud		2.292	2.282	2.222	2.145	2.217	2.261	2.195	2.234	2.305	2.345	2.340	2.324
	Ebn at Noon		215	243	255	256	265	267	256	252	248	230	201	196
	Edh at Noon		21	28	36	43	42	41	43	39	33	26	20	17
All-Sky Solar Radiation	RadAvg		228	448	824	1317	1634	1714	1601	1426	1022	573	262	171
	RadStd		41	80	103	174	177	160	188	122	137	83	36	21

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+45
(47)	Regional (25 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+43

Nomenclature: See separate page